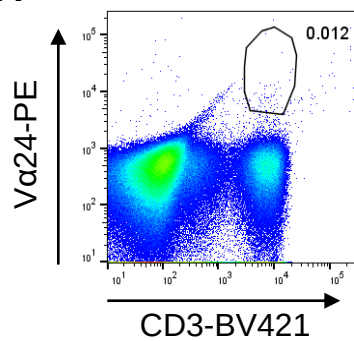


Anti-GD2 CAR-NKT cells in relapsed or refractory neuroblastoma: updated phase 1 trial interim results

In the format provided by the
authors and unedited

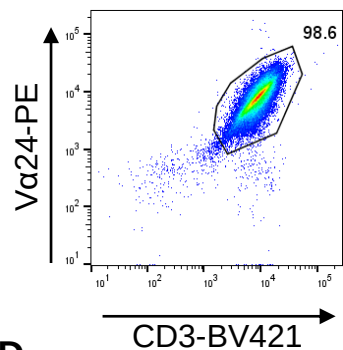
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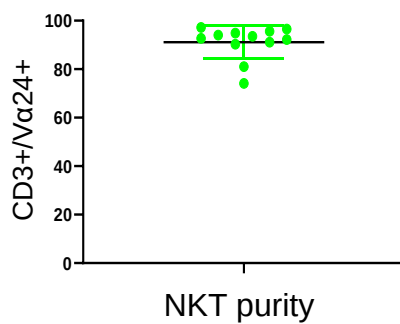


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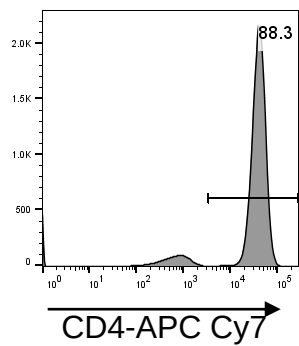
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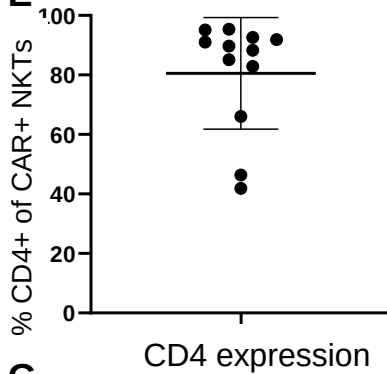
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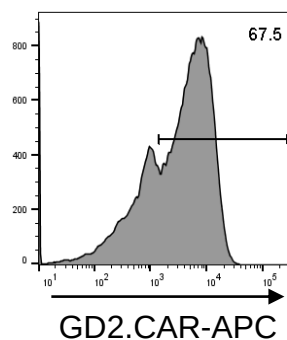
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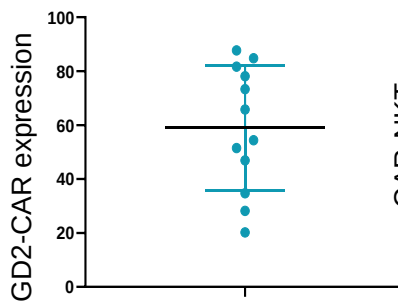
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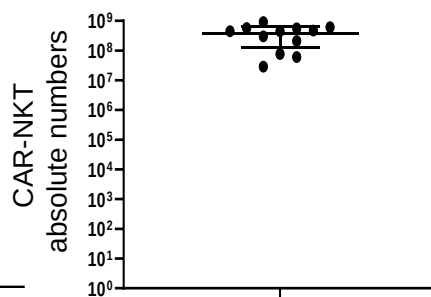
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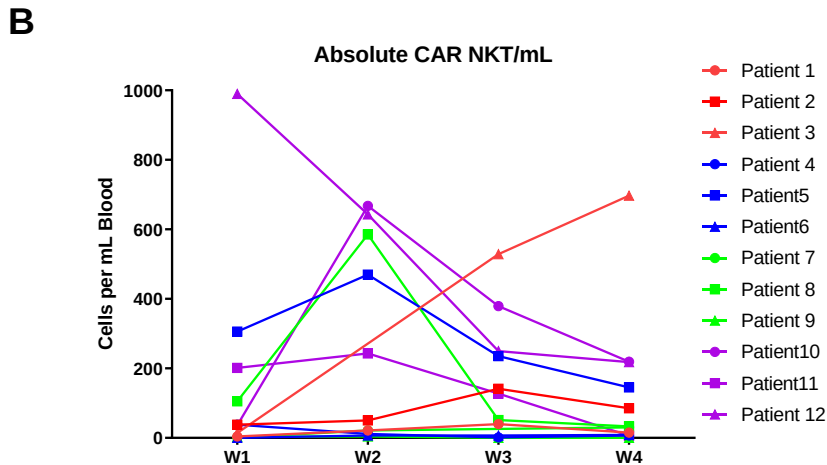
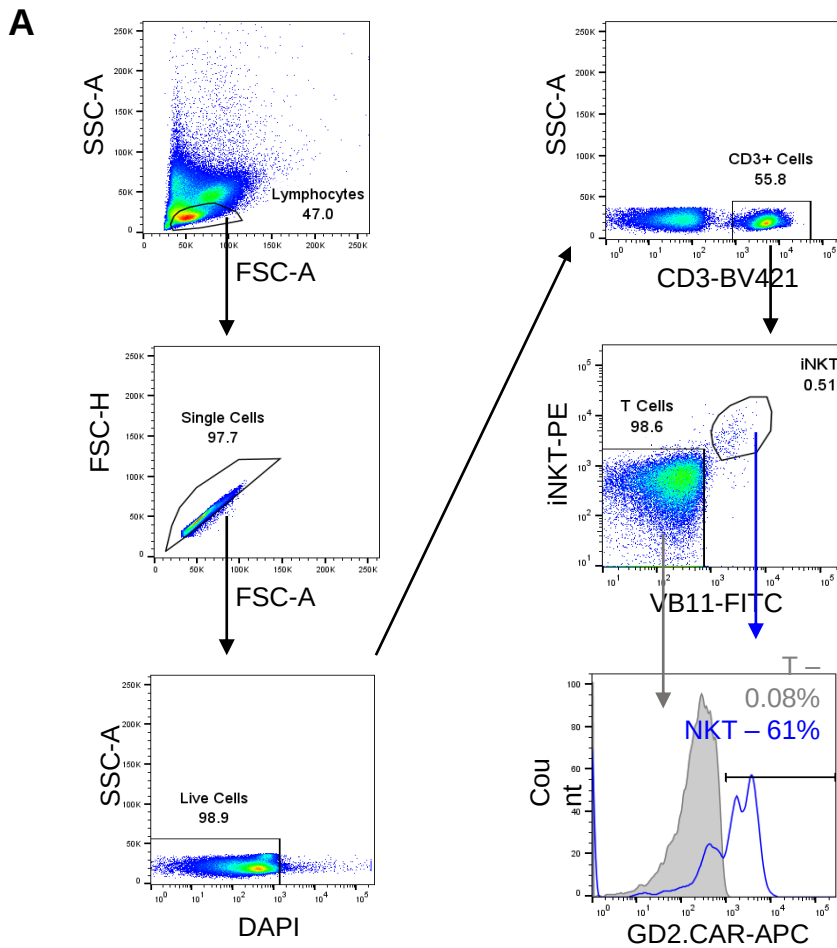
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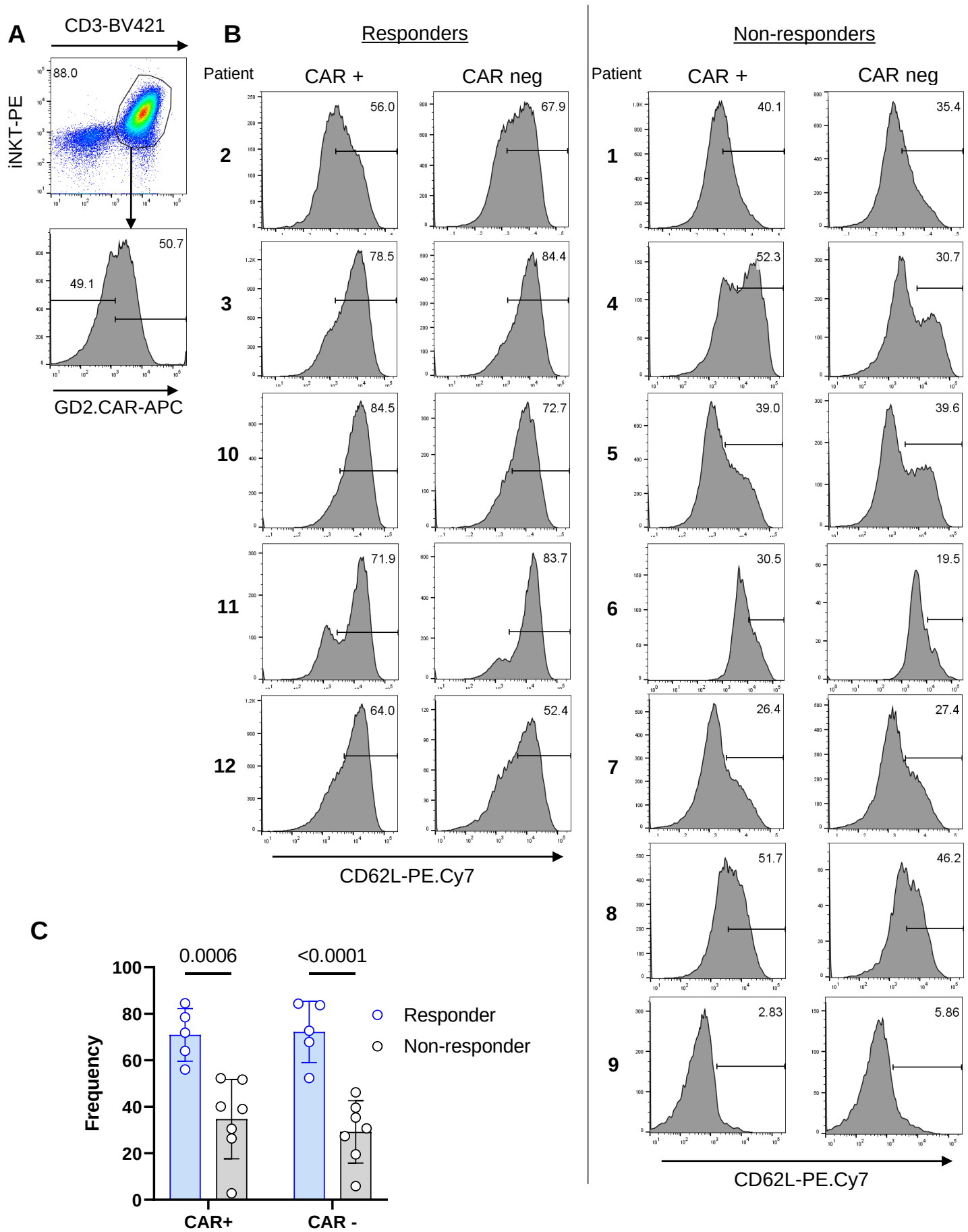
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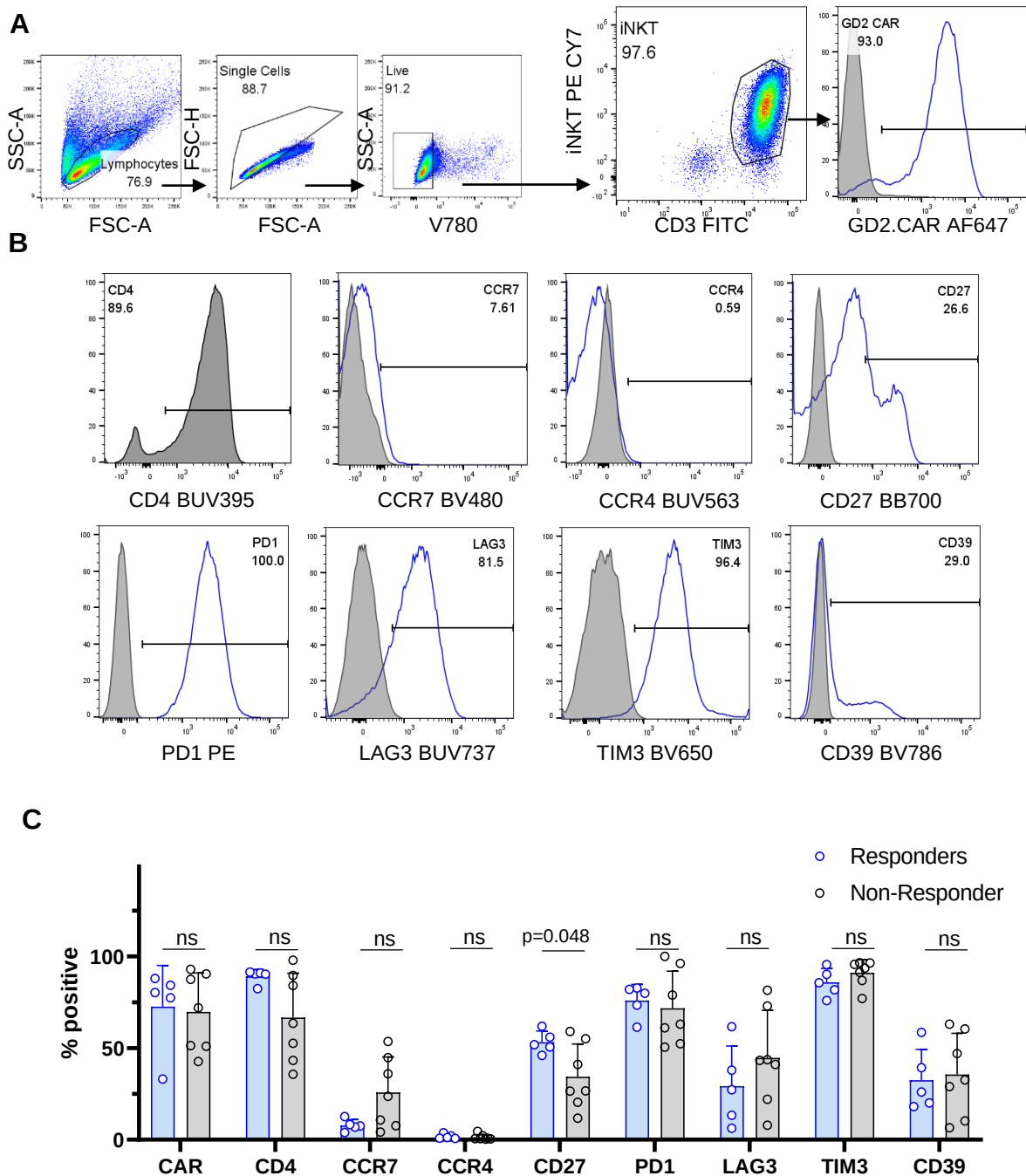
Supplementary Figure 1. GD2-CAR.15 NKT product manufacturing parameters. NKTs were selected with iNKT-specific microbeads from peripheral blood apheresis products, stimulated with alpha-galactosylceramide-loaded, irradiated peripheral blood mononuclear cells, and transduced with a gamma-retroviral vector encoding the GD2-CAR and IL15 transgenes. CAR-NKT cell products were assessed in a GMP FACS laboratory. **A.** NKT frequency in a representative apheresis product. NKT purity in pre-infusion products for **(B)** a representative patient and **(C)** all patients, n=12, data shown as mean +/- SD. CD4 frequency in pre-infusion products for **(D)** a representative patient and **(E)** all patients, n=12, data shown as mean +/- SD. CAR expression in pre-infusion products for **(F)** a representative patient and **(G)** all patients, n=12, data shown as mean +/- SD. **H.** GD2-CAR.15 NKT absolute numbers in pre-infusion products from all patients, n=12, data shown as mean +/- SD.



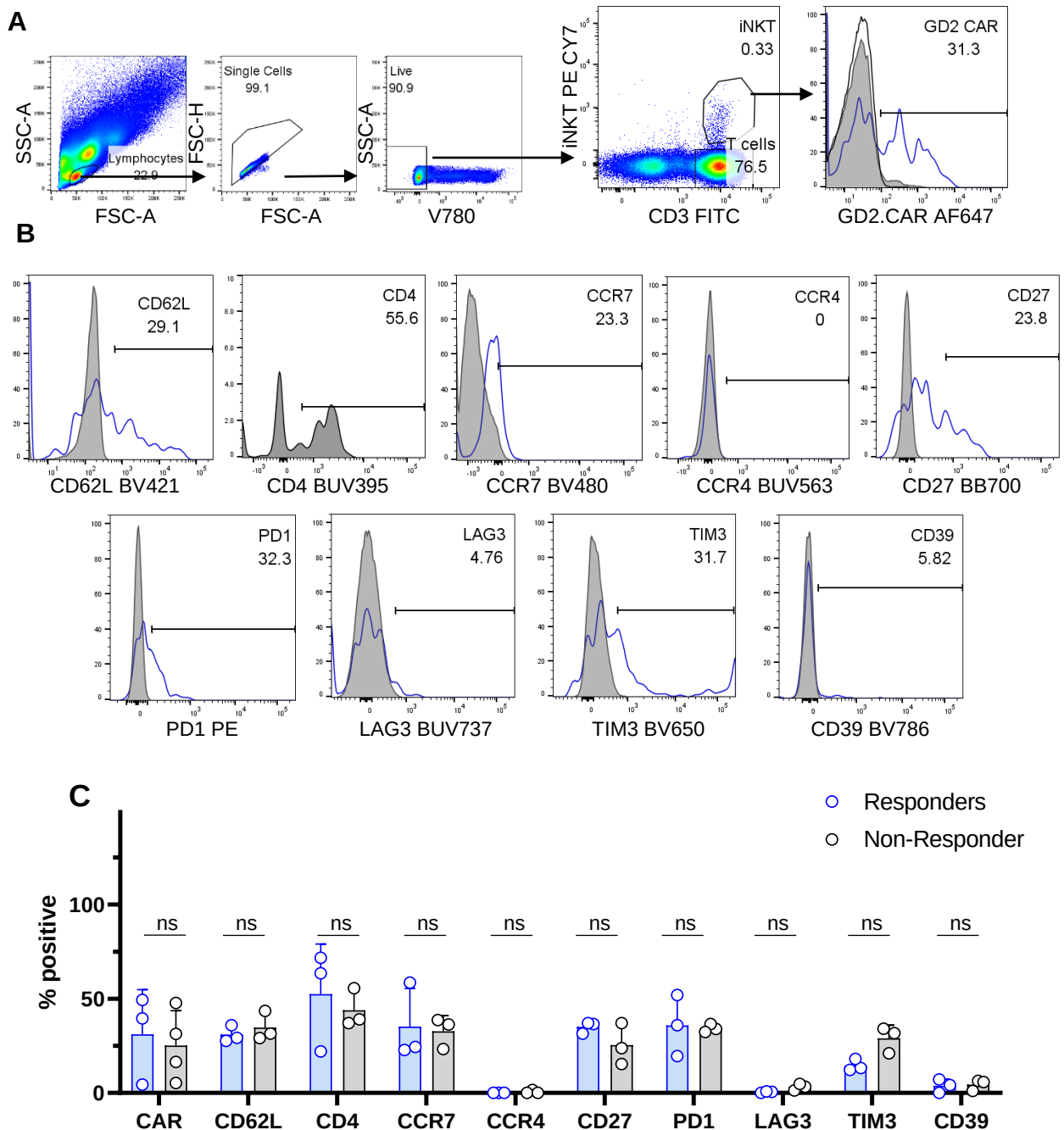
Supplementary Figure 2. Quantifying GD2-CAR.15 NKTs in peripheral blood and tumor biopsy samples. **A.** Patient-derived samples were processed and stained with indicated fluorochrome-conjugated antibodies. After gating on live cells and removing duplets, the iNKT population was defined as CD3⁺iNKT⁺Vβ11⁺. The GD2-CAR NKT population was identified using the iNKT-Vβ11⁻ population as an internal control. **B.** Peripheral blood samples were collected at indicated time points and the CAR transgene was detected by flow cytometry post-infusion. Shown are absolute numbers of circulating CAR⁺ NKT cells.



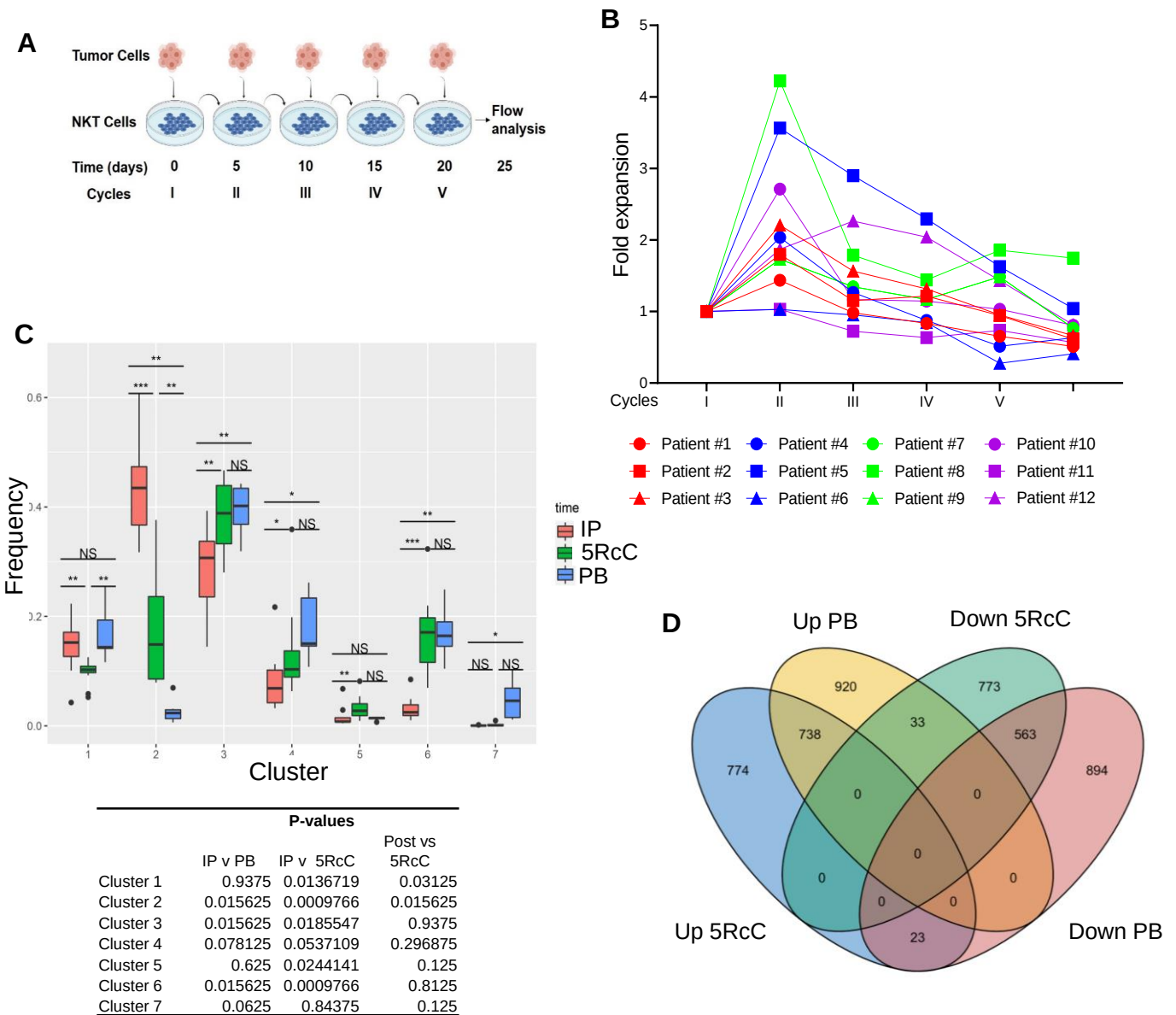
Supplementary Figure 3. CD62L expression in infusion products. Patient products were stained for NKT markers, CAR, and CD62L expression. **A.** Representative example of gating strategy. **B.** CD62L expression in CAR^{+/−} subsets of responder and non-responder infusion products. **C.** Comparison of CD62L expression in CAR^{+/−} NKTs of responder (n=5) versus non-responder (n=7) groups. Data shown as mean $\pm$ SD. Groups were compared using two-way ANOVA with Sidak's correction for multiple comparisons.



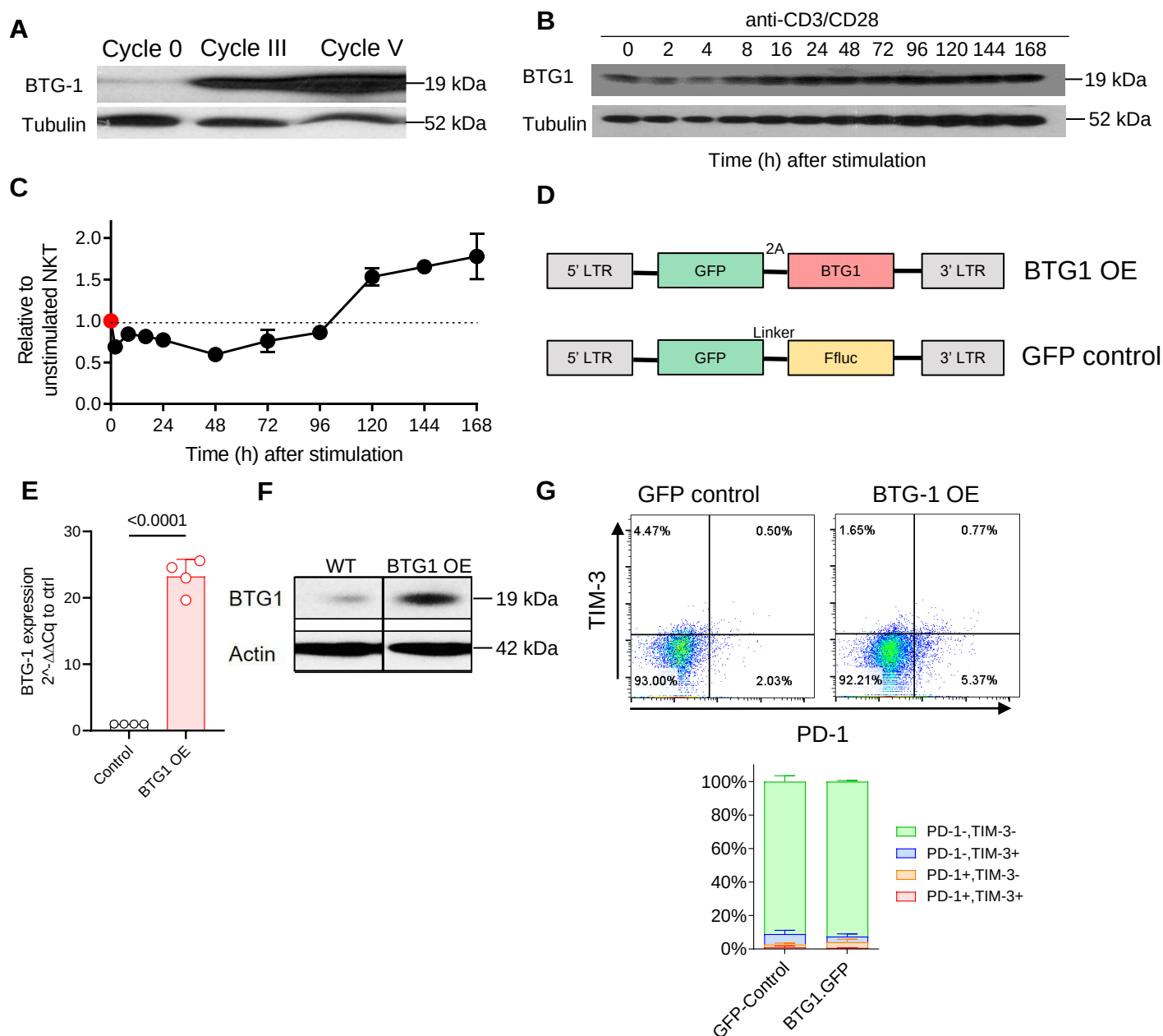
Supplementary Figure 4. Phenotype of GD2-CAR.15 NKT products. Manufactured products were thawed and rested overnight in complete medium containing IL2. Cells were stained for surface expression of exhaustion/activation and memory markers. **A.** Representative flow cytometry plots and gating strategy. **B.** Representative histograms of exhaustion/activation and memory marker expression in CAR⁺ NKT cells. **C.** Summary of exhaustion/activation and memory marker expression in all 12 infusion products. Products were separated into responders (n=5) versus non-responders (n=7) data shown is mean +/- SD and groups were compared using a two-tailed unpaired T test.



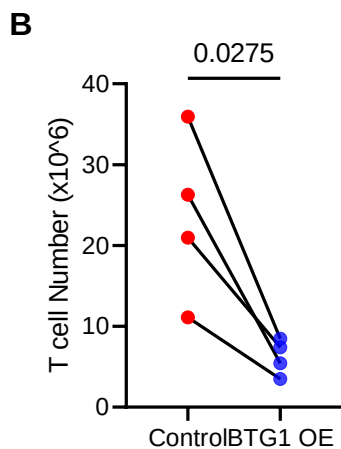
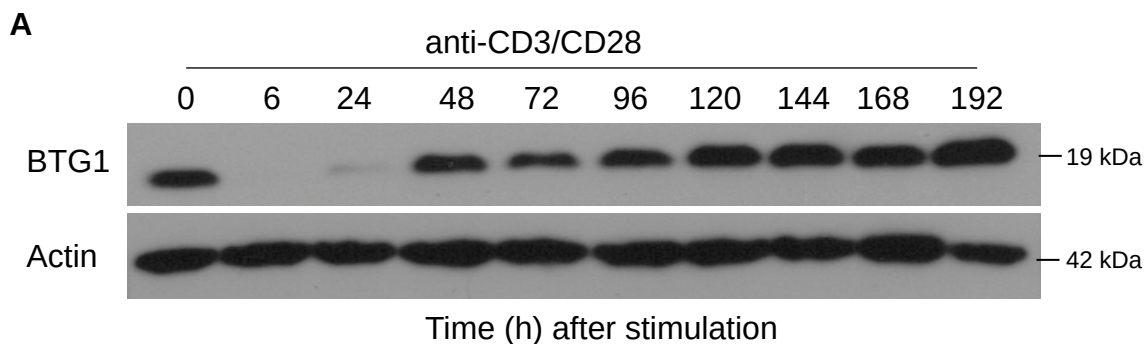
Supplementary Figure 5. GD2-CAR NKT phenotype in peripheral blood post-infusion. PBMCs isolated from patient peripheral blood 2-3 weeks post-infusion were thawed and rested overnight in complete medium with IL2. Cells were stained for surface expression of exhaustion/activation and memory markers. **A.** Representative flow cytometry plots and gating strategy to identify GD2-CAR⁺ NKTs. **B.** Representative histograms of exhaustion/activation and memory markers in CAR⁺ NKT cells. **C.** Summary of exhaustion/activation and memory markers in all 12 infusion products. Products were separated into responders (n=5) versus non-responders (n=7), data shown is mean $\pm$ SD and groups were compared using a two-tailed unpaired T test.



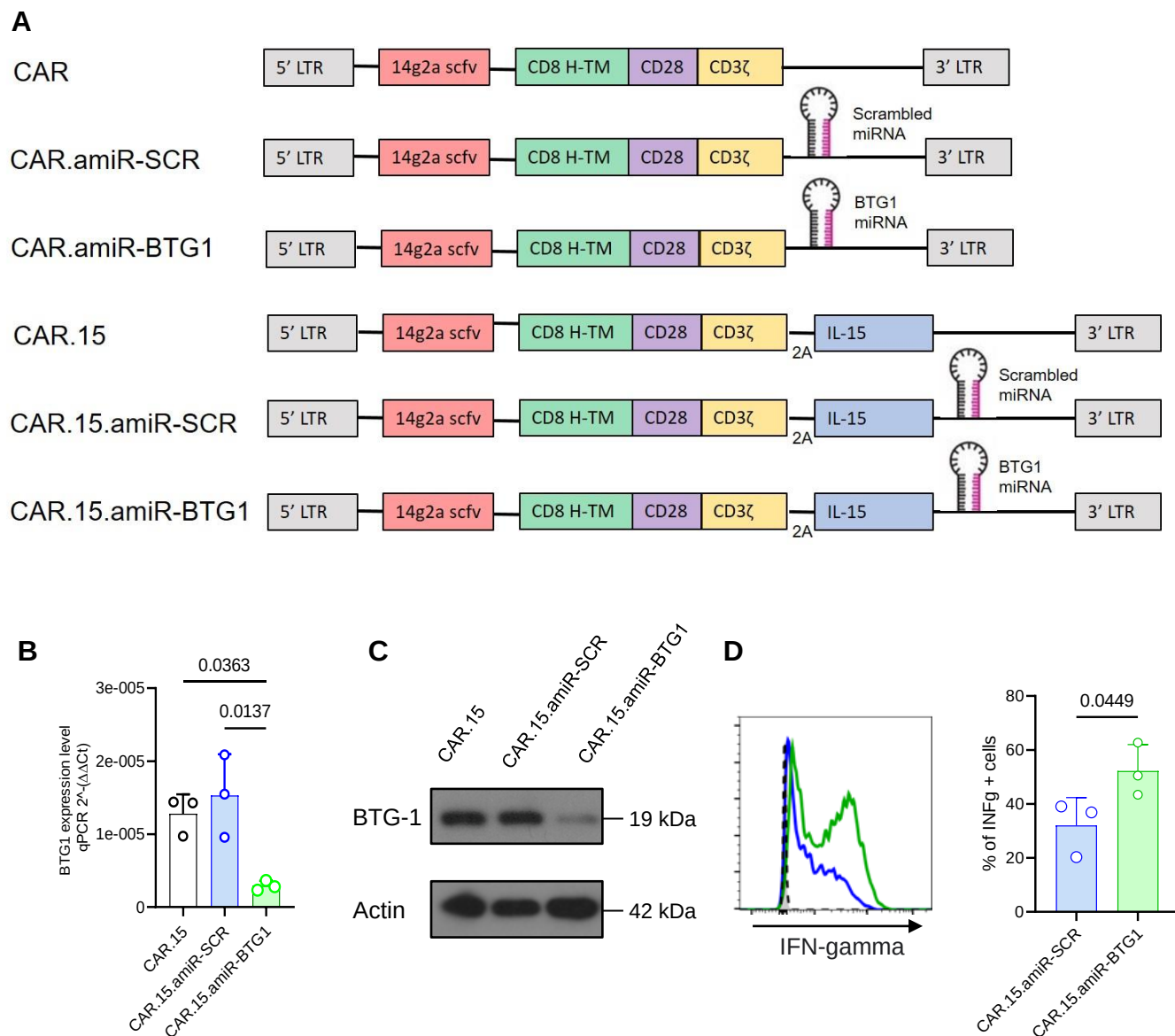
Supplementary Figure 7. Functional and transcriptomic characterization of GD2-CAR.15 NKTs. **A.** Design of repeat tumor co-culture system to induce GD2-CAR.15 NKT exhaustion via chronic antigen exposure. CAR-NKTs are re-plated with fresh CHLA255 NB tumor cells every five days for multiple cycles. **B.** Fold expansion of pre-infusion product CAR-NKTs from baseline over the course of five repeat co-culture cycles with CHLA255 cells corresponding to results shown in **Extended Data Figure 6A**. **C.** Frequencies of cells from pre-infusion products (IP)(n=12), five-cycle repeat co-culture with tumor cells (5RcC)(n=12), and CAR-NKTs isolated from peripheral blood (PB)(n=7) post-infusion in each of seven gene expression clusters from **Supplementary Figure 6**. Box plots show the median in the center. The lower and upper quartile of the box show the 25th and 75th percentiles, respectively. The lower and upper whiskers show the 5th and the 95th percentiles, respectively, black dots represent outliers, defined as data beyond the ± 1.5 x interquartile range. Clusters were compared using Wilcoxon test and p values for each comparison are shown. **D.** Venn diagram demonstrating shared and unshared differentially expressed genes in GD2-CAR.15 NKTs post-infusion in PB and after 5RcC.



Supplementary Figure 8. BTG1 expression is elevated in serially stimulated CAR-NKTs and increases over time in TCR-stimulated NKTs. **A.** BTG1 protein expression in NKT cells as determined by Western blot at indicated cycles in repeat tumor challenge assay, one representative of two donors. Unprocessed blot data can be found in Supplementary Figure 11. **B.** BTG1 protein expression in NKT cells following activation with CD3/CD28-specific monoclonal antibodies. Experiment was performed once. Unprocessed blot data can be found in Supplementary Figure 11. **C.** BTG1 expression in NKT cells as determined by qPCR after activation with CD3/CD28 specific monoclonal antibodies relative to unstimulated NKts, n=3 technical replicates, data shown is mean $\pm$ SD. **D.** Design of BTG1 overexpression (BTG OE) construct and GFP control construct for expression in NKts. **E.** BTG1 gene expression as determined by qPCR, n=4 donors, data shown is mean $\pm$ SD, two-tailed unpaired T test, and **(F)** protein expression as determined by Western blot in NKts expressing the BTG1 OE construct versus control NKts, one representative of two donors. Unprocessed blot data can be found in Supplementary Figure 11. **G.** Representative plots showing expression of exhaustion markers TIM-3 and PD-1 in BTG1-GFP NKts and GFP control NKts (top) and distribution of TIM-3⁺/ PD-1⁺ populations in BTG1-GFP NKts and GFP control NKts (bottom), (n=3). Data shown is mean $\pm$ SD.



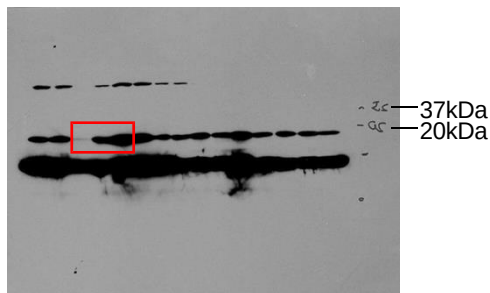
Supplementary Figure 9. BTG1 expression increases during prolonged T cell receptor stimulation and transgenic BTG1 overexpression inhibits T cell expansion. Freshly isolated naïve T cells (CD45RA⁺CCR7⁺) were stimulated with plate-bound CD3/CD28 monoclonal antibodies and cultured in the presence of IL2. **A.** BTG1 expression in T cells measured by Western blot at indicated timepoints, one representative of two donors. Unprocessed blot data can be found in Supplementary Figure 11. **B.** Absolute number of T cells expressing the BTG1 OE construct or GFP control following stimulation (0.5×10^6 cells) and *ex vivo* culture, $n=4$. Data shown is mean $\pm$ SD, two-tailed paired T test.



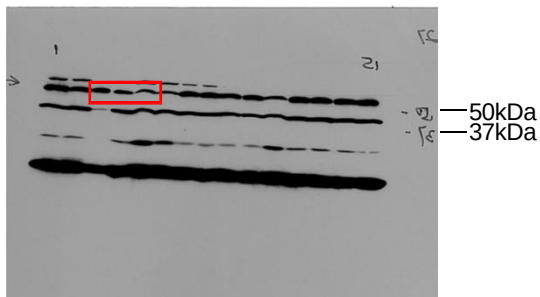
Supplementary Figure 10. BTG1-targeting amiR constructs downregulate BTG1 expression in GD2-CAR.15 NKTs. **A.** Retroviral constructs for *BTG1* KD. Constructs encode the GD2-CAR with and without IL15, each with an artificial microRNA targeting *BTG1* or a scrambled control. amiRs were cloned into MMuLV-based gamma-retroviral constructs downstream from the GD2-CAR. **B.** *BTG1* gene expression quantified by qPCR, (n=3). Data shown is mean +/- SD, Ordinary one-way ANOVA and **(C)** *BTG1* protein expression by Western blot in NKTs expressing indicated constructs, one representative of two donors. Unprocessed blot data can be found in Supplementary Figure 11. **D.** CAR-NKT cells were stimulated with anti-CD3/CD28 beads followed by stimulation with PMA/ionomycin. Intracellular staining for IFN γ was performed following fixation and permeabilization. A representative histogram showing IFN γ expression in NKTs expressing indicated constructs and summary data of IFN γ expression, (n=3). Data shown is mean +/- SD, two-tailed paired T test.

Supplementary Figure 8A.

BTG1

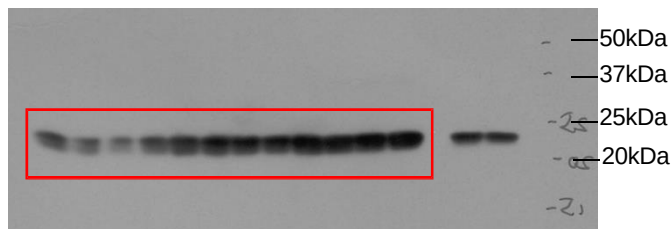


Tubulin

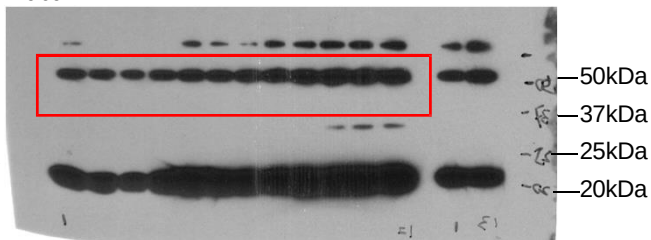


Supplementary Figure 8B.

BTG1

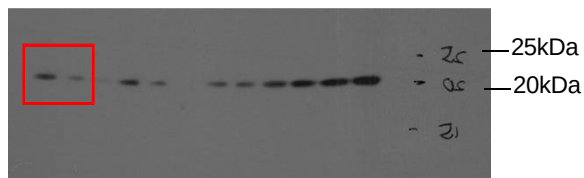


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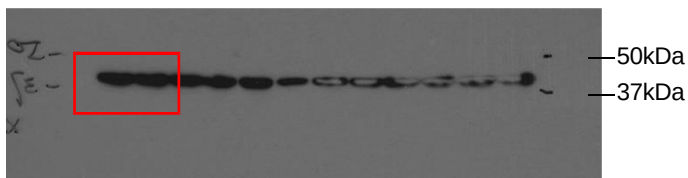


Supplementary Figure 8F.

BTG1

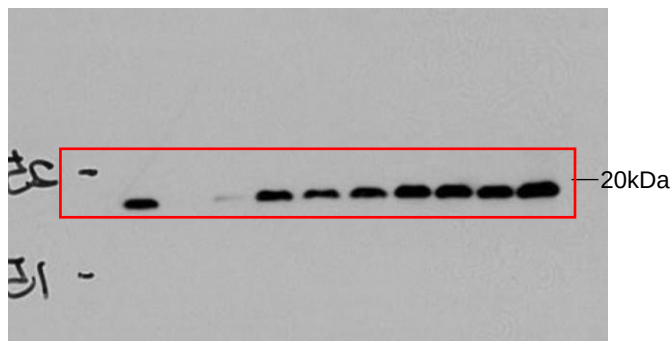


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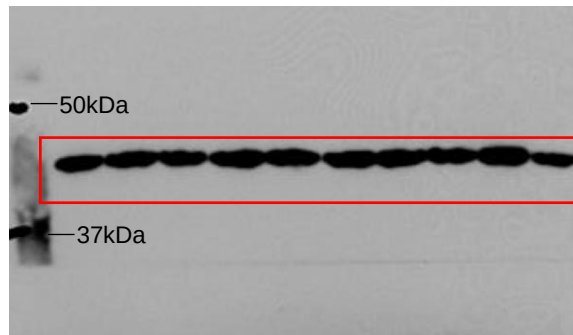


Supplementary Figure 9A.

BTG1

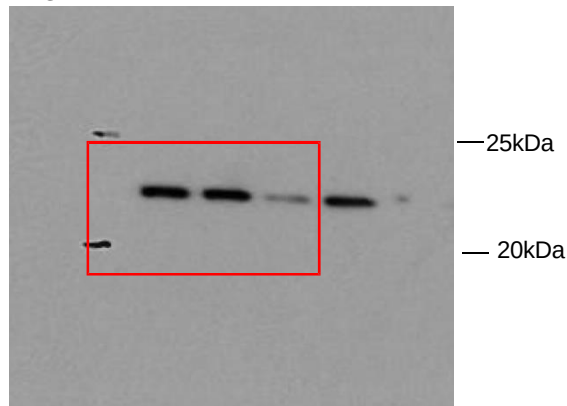


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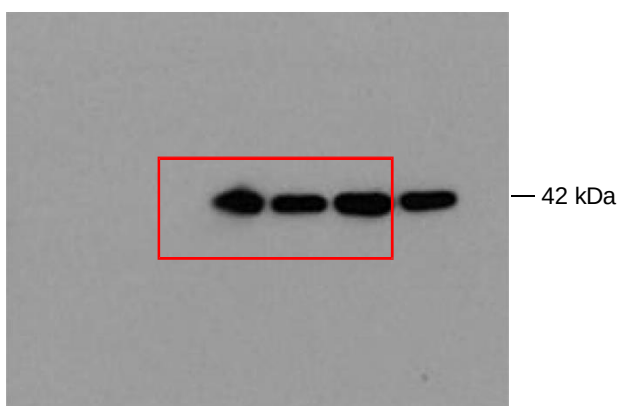


Supplementary Figure 10C.

BTG1



Actin



Supplementary Figure 11. Unprocessed western blot data for Supplementary Figures 8, 9, 10.